

Tableau Server Maintenance Essentials

by Mohamad Ikhsan Nurulloh

Introduction to Tableau Server



Tableau Server is a business intelligence and data visualization platform that allows users to share, distribute, and manage Tableau workbooks, dashboards, and data sources securely across their organization.

<https://www.tableau.com/products/server>

Why Tableau Server Needs Maintenance

- **Regular Backups:** Creating backups is essential to protect data and ensure recovery in case of failures.
- **File Management:** Removing unneeded files prevents storage issues and optimizes server performance.
- **Performance Monitoring:** Regularly checking server status and analyzing activity helps identify potential issues before they escalate.
- **Upgrades:** Regularly upgrading Tableau Server ensures access to the latest features, performance improvements, and security patches.

Tableau Server

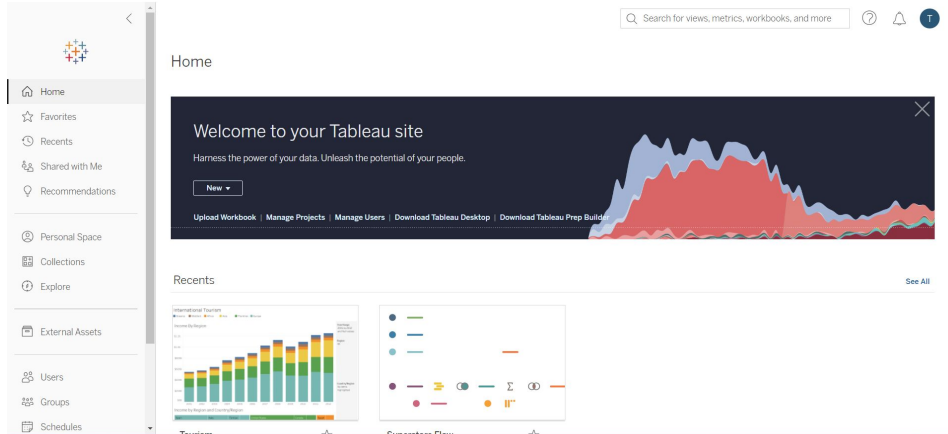


Tableau Server

 + a b l e a u

STATUS

MAINTENANCE

CONFIGURATION

Tableau Server is running  [sign out](#)

Process	node1	node2	node3
Gateway			
Application Server			
VizQL Server			
Cache Server			
Cluster Controller			
Search & Browse			
Backgrounder			
Data Server			
Data Engine			
File Store			
Repository			
TSM Controller			
License Server			

Refresh Status

Active

Busy

Error

Stopped

Tableau Service Manager (TSM)

Tableau Server

Tableau Server

Server Status

Process Status

The real-time status of processes running in Tableau Server.

Process	tableau-server
Cluster Controller	✓
Gateway	✓
Application Server	✓
VizQL Server	✓
Cache Server	✓
Search & Browse	✓
Backgrounder	✓
Data Server	✓
Data Engine	✓
File Store	✓
Repository	✓
Tableau Prep Conductor	✓
Metrics	✓

✓ Active ⌂ Busy ✓ Passive ⚠ Unlicensed ✗ Down ⚙ External □ Status unavailable

The Tableau Server UI provides access to published dashboards, manages job extraction, and oversees user activity and permissions.

Tableau Server



- Home
- Favorites
- Recents
- Shared with Me
- Recommendations

- Personal Space
- Collections
- Explore

- External Assets

- Users
- Groups
- Schedules
- Jobs
- Tasks
- Server Status**
- Settings

Server Status

Analysis

Dashboard	Analysis
Traffic to Views	Usage and users for published views.
Traffic to Data Sources	Usage and users for published data sources.
Actions by All Users	Actions for all users.
Actions by Specific User	Actions for a specific user, including items used.
Actions by Recent Users	Recent actions by users, including last action time and idle time.
Background Tasks for Extracts	Completed and pending task details for extract refresh.
Background Tasks for Non Extracts	Completed and pending background task details for non-extract refresh.
Flow Performance History	View flow run performance.
Stats for Load Times	View load times and performance history.
Stats for Space Usage	Space used by published content, including extracts and live connections.
Backgrounder Dashboard	Backgrounder job runtimes and busy periods.
Background Task Delays	Difference between scheduled and actual start times of background tasks.
Performance of Views	Overall distribution of view load times and slowest views in a given time period.
Server Disk Space	Current and historical disk space usage, by server node.
Stale Content	Age and usage details for content that hasn't been recently used.
Ask Data Usage	Ask Data usage and adoption details.
Data Quality Warning History	Data quality warnings in use, edited, or removed.

Tableau Service Manager (TSM)

Tableau Service Manager (TSM) Web UI

+ tableau		STATUS	MAINTENANCE	CONFIGURATION	Tableau Server is running	sign out
Process	node1 localhost					
Gateway	✓					
Application Server	✓					
Interactive Microservice Container	✓					
VizQL Server	✓					
Cache Server	✓					
Cluster Controller	✓					
Search & Browse	✓					
Backgrounder	✓					
Non-Interactive Microservice Container	✓					
Data Server	✓					
Data Engine	✓					
File Store	✓					
Repository	✓					
Tableau Prep Conductor	✓					
Tableau Prep Flow Authoring	✓					
Tableau Prep Minerva Service	✓					
Ask Data	✓					
Metrics Service	✓					
Messaging Service	✓					
Data Source Properties Service	✓					
Internal Data Source Properties Service	✓					
TSM Controller	✓					
License Server	✓					
Activation Service	✓					
Content Exploration Service	✓					
Collections Service	✓					
Statistical Service	✓					
Query Gateway Service	✓					
Data Profiling Service	✓					
Query Policy Service	✓					
Virtual Connections Service	✓					
Extract Service	✓					
Index And Search Server	✓					
Non-Relational Storage Service	✓					
Data Stories	✓					
Refresh Status		✓ Active	⚙ Busy	⚠ Passive	⚠ Unlicensed	⚠ Degraded
		⚠ Error	⚠ External	⚠ Stopped	⚠ Status Unavailable	

Single Node

+ tableau		STATUS	MAINTENANCE	CONFIGURATION	Tableau Server is running	sign out
Process	node1	node2	node3			
Gateway	✓	✓	✓			
Application Server	✓	✓	✓			
VizQL Server	✓✓✓✓	✓✓✓✓	✓✓✓✓			
Cache Server	✓✓	✓✓	✓✓			
Cluster Controller	✓	✓	✓			
Search & Browse	✓	✓	✓			
Backgrounder	✓✓	✓✓	✓✓			
Data Server	✓✓	✓✓	✓✓			
Data Engine	✓	✓	✓			
File Store	✓	✓	✓			
Repository	✓		✓			
TSM Controller	✓					
License Server	✓					
Refresh Status				✓ Active	⚙ Busy	⚠ Error
				⚠ Stopped		

Multi Node (Distribute Compute) - High Availability (HA)

The TSM web UI provides a user-friendly interface for managing Tableau Server configurations and performing various administrative tasks.

Tableau Service Manager (TSM) Web UI

[STATUS](#)[MAINTENANCE](#)[CONFIGURATION](#)[Tableau Server is running](#) ▼[sign out](#)

Server Maintenance

Maintain and improve Tableau Server performance by generating and analyzing log files, and performing other maintenance tasks.

Log Files

Generate a custom log file snapshot. Generating a log file snapshot may take several minutes. After Tableau Server has generated the snapshot, you can download the snapshot or upload it to Tableau Technical Support. Ensure that you have an open *Support Case* before uploading logs. [Learn more](#)

[Generate Log File Snapshot](#)[Upload To Technical Support Case](#)[Download](#)[Delete](#)

Description	Created	Range	Size	Stored in	Status
-------------	---------	-------	------	-----------	--------

Server Crash Reporter

Send logs and related files to Tableau when the server has an issue that results in a crash. Do not enable this feature if your organization's data is subject to any privacy regulations.

[Learn more](#)☐ Enable crash reporting

Daily at 01:00 UTC ▼

Product Usage Data

Help us improve your Tableau experience by sharing how you use the product. Tableau collects information about your feature usage. All usage data will be handled according to our Privacy Policy - <http://www.tableau.com/privacy>

[Learn more](#)☒ Don't send product usage data

Use the tsm command-line utility to schedule backups and perform other advanced maintenance operations. [Learn more](#)

Tableau Service Manager (TSM) Web UI

 **tableau**

STATUS

MAINTENANCE

CONFIGURATION

Tableau Server is running  [sign out](#)

Topology

Security

User Identity & Access

Notifications

Licensing

Topology

Configure and improve Tableau Server performance by adding or removing nodes and changing process configurations and other settings. [Learn more](#)

node1
tableau-server

Gateway

☒

Application Server

1

▼

Interactive Microservic...

1

▼

VizQL Server

1

▼

Cache Server

1

▼

Cluster Controller

☒

Search & Browse

☒

Backgrounder

1

▼

Non-Interactive Micros...

1

▼

Data Server

1

▼

Data Engine

☒

File Store

☒

Repository

☒

Tableau Prep Conductor

☒

Add a Node

Step 1

Download the node bootstrap configuration file and locate your Tableau Server installer. The same installer can be used to install multiple nodes.
[Having trouble finding the installer?](#)

☒ Include temporary credentials in file

Download Bootstrap File

Step 2

Run the node installer on the new node, and when prompted, provide the configuration file.

Tableau Services Manager will detect the new node and display it on the Topology page.

[Learn more about adding, removing, and managing nodes in Tableau Services Manager.](#)

Tableau Service Manager (TSM) CLI



SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE



```
Swap usage:  0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

10 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 25 12:48:21 2024 from 35.235.244.50
ikhsannur1996@tableau-server:~$ tsm
Tableau Services Manager command line Utility -- 20231.24.0718.1427

tsm help                -- Help for tsm commands.
tsm help <category>     -- Show help for a specific command category.
tsm help [category] <command> -- Show help for a specific command.
tsm help commands       -- List all top-level commands/categories.

tsm help [parameters]

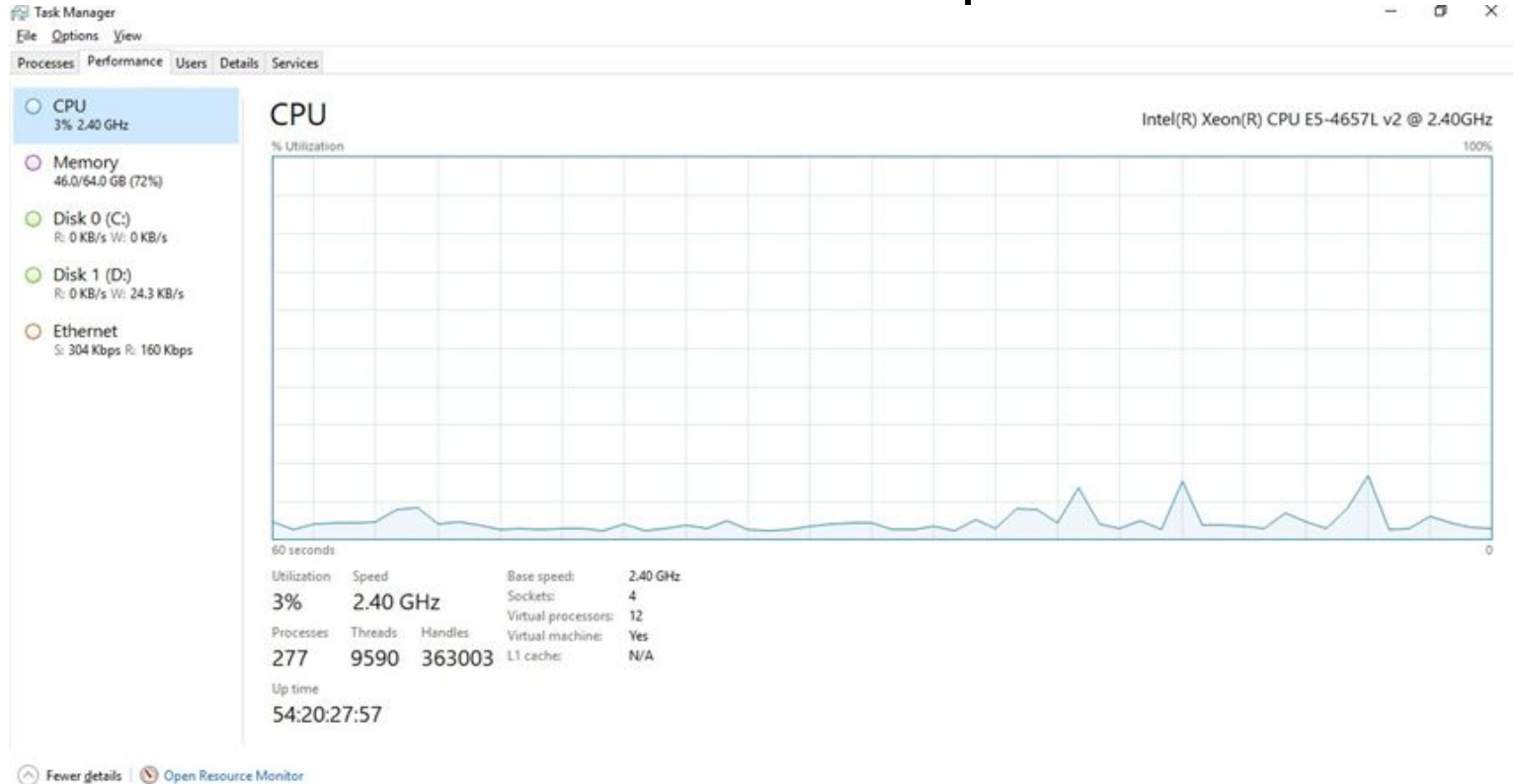
Global Parameters
-h,--help                Display TSM help.
-p,--password <PASSWORD> Use the specified password.
-s,--server <URL>        Use the specified Tableau Services
                          Manager URL. If no protocol is
                          specified, https://<localhost
                          dnsname>:8850 is assumed.
--trust-admin-controller-cert Do not validate the server certificate.
-u,--username <USER>     Use the specified user name.

Example
tsm help
ikhsannur1996@tableau-server:~$
```

The TSM CLI offers a more advanced way to manage Tableau Server using command-line scripts, which can be useful for automating repetitive tasks.

Tableau Server Health Check

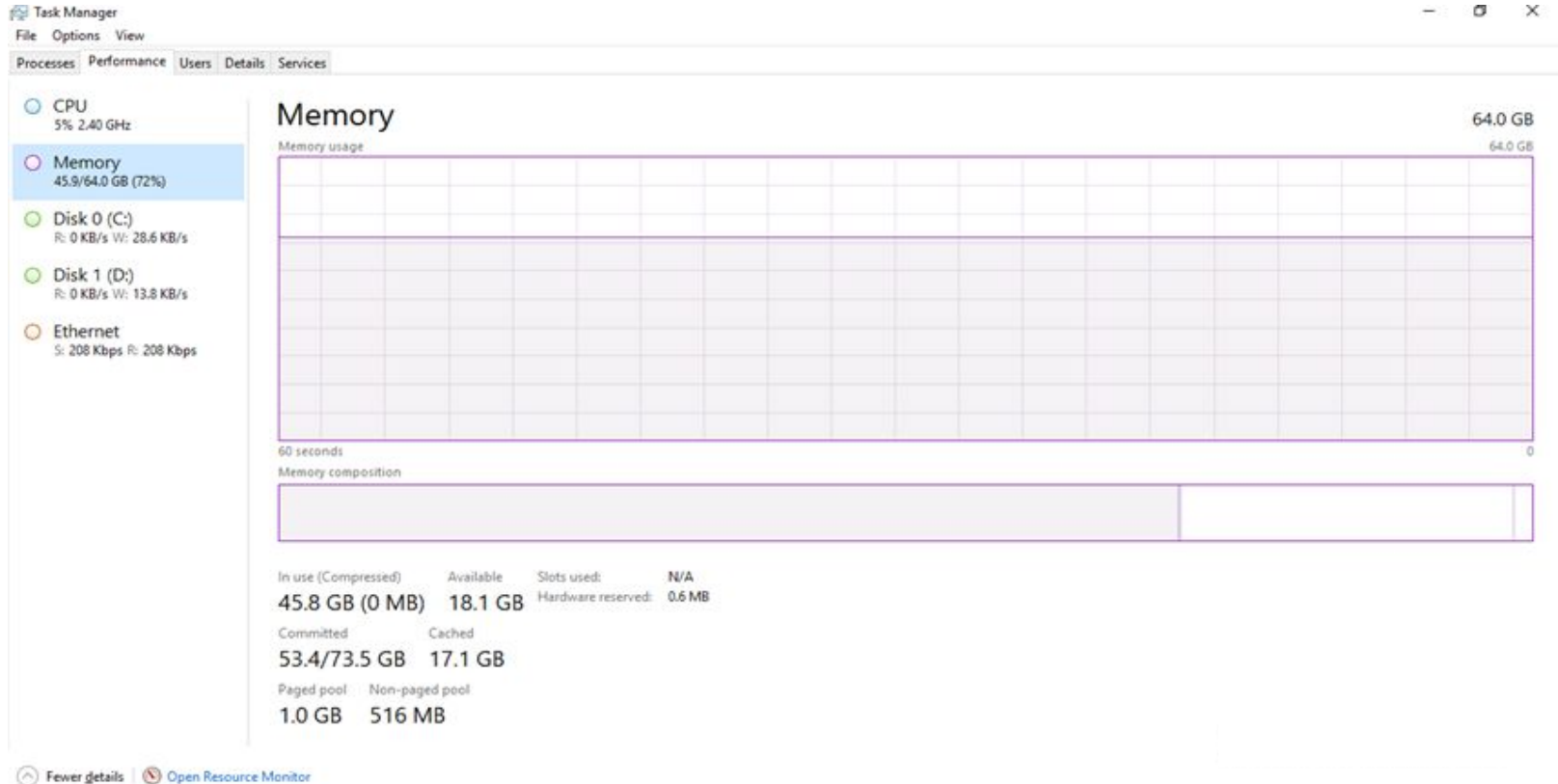
Server Health Check Up - CPU Load



Monitoring CPU Load: Regularly monitor CPU usage to ensure it does not exceed safe limits.

Optimization: Provide tips on optimizing CPU usage, such as upgrading hardware or optimizing queries.

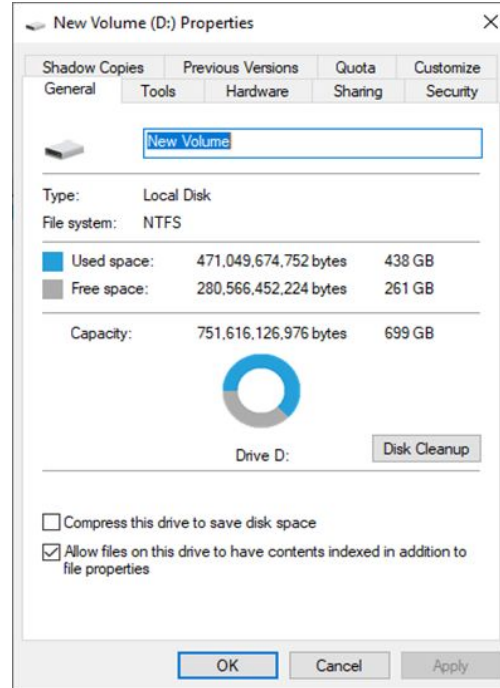
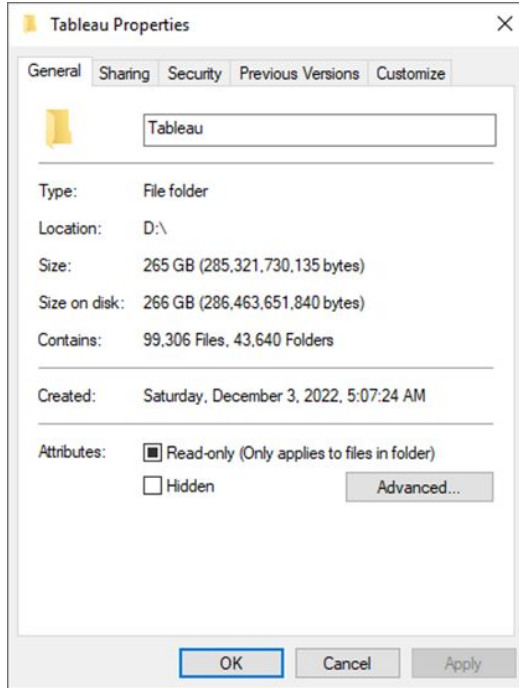
Server Health Check Up - Memory Load



Monitoring Memory Usage: Regularly check memory usage to prevent out-of-memory errors.

Optimization: Provide tips on optimizing memory usage, such as upgrading RAM or optimizing application settings.

Server Health Check Up - Free Disk



Monitoring Disk Space: Regularly check available disk space to prevent storage issues.

Optimization: Provide tips on optimizing disk space usage, such as deleting unnecessary files or upgrading storage

Devices and drives (3)



Local Disk (C:)

62.4 GB free of 99.4 GB



New Volume (D:)

261 GB free of 699 GB

Tableau Server Maintenance Essentials

Logshark Analysis

Logshark Forum



Get Started

[Get your computer set up for LogShark](#)

[Install LogShark](#)

Use Logshark

[Run LogShark](#)

[Configure and Customize LogShark](#)

[Publish Results to Tableau Server](#)

LogShark Installation and User Guide

[Edit this page](#) [Submit an issue](#)

Analyze your Tableau log files – in Tableau!

LogShark is a tool you can use to analyze and troubleshoot Tableau performance and activity. LogShark extracts data from Tableau Server and Tableau Desktop log files and builds workbooks that can help you identify and understand error conditions, performance bottlenecks, and background activity.

LogShark works by running a set of targeted plugins that pull specific data out of the log files. LogShark builds a data source and provides Tableau workbooks which you can then use to analyze the log files in Tableau.

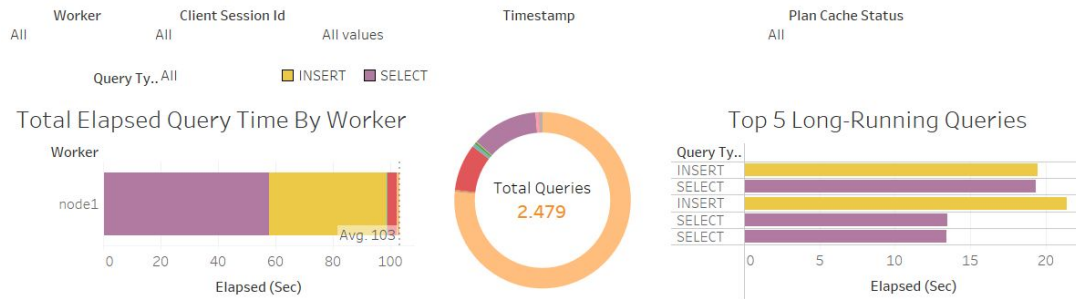
NOTE: This version of LogShark is significantly faster and brings back the ability to write to a PostgreSQL database server, as well as a number of other improvements. See [releases page](#) for a full list of updates.

This installation and user guide will walk you through the steps to get started, how to run LogShark and view the results, and a command dictionary.

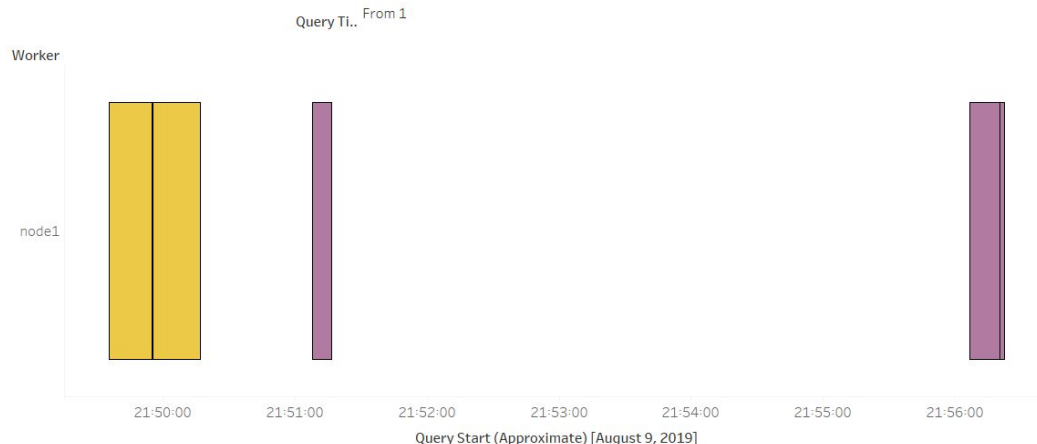
Sample Logshark Analysis - Hyper Query

Hyper Query Timeline

Displays query-end events from the Hyper logs. This can be used to identify slow queries.
Use the **Query Time Threshold** filter to display only queries exceeding the given threshold...



Query Timeline By Worker..



Hyper Query Analysis:

Explain how to use Logshark to analyze hyper queries, which are complex queries that can impact performance.

Logshark Analysis - Server Resource Manager

Server Resource Manager Reported Total Utilization

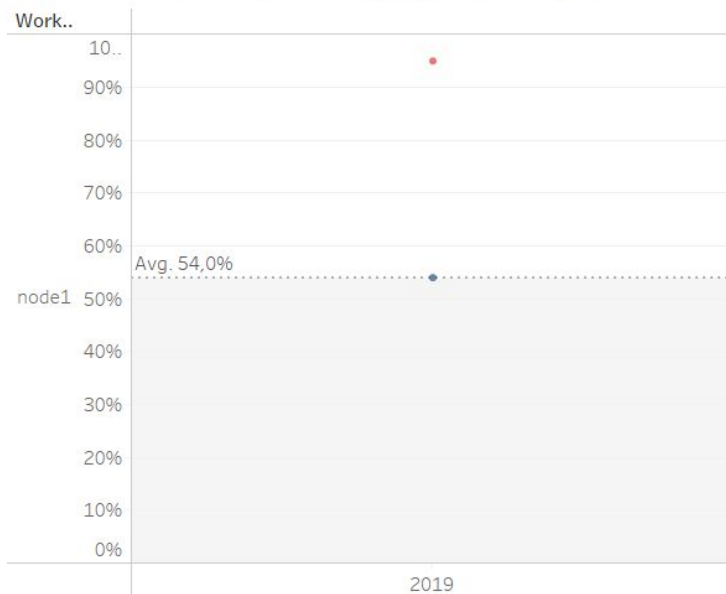
This dashboard shows Tableau Server Resource Manager logging activity on the input logset. The red line represents a configured global resource threshold. There are multiple reasons why a performance issue may not be visible here. Only a few processes are actively monitored by Tableau Server. Log files may sometimes fail to be collected during periods of performance degradation. Additionally, the customer may have reconfigured or disabled SRM entirely, or be using a product vers...

All values

Timestamp

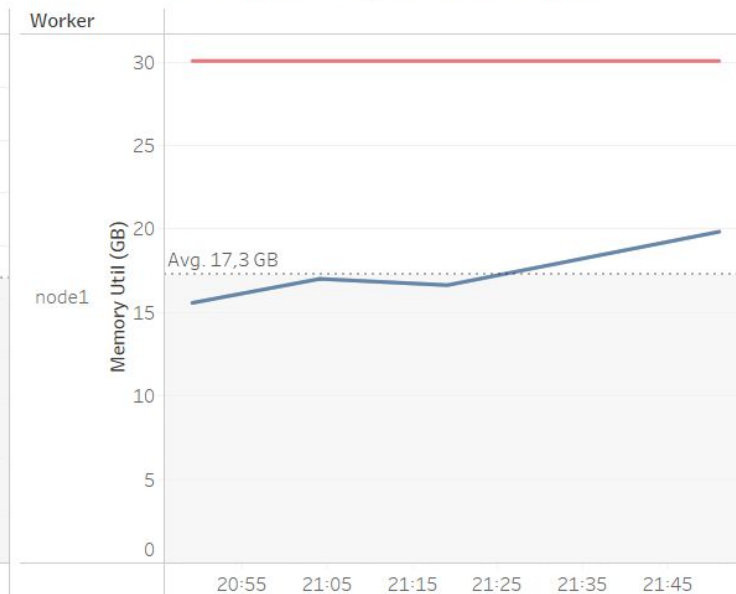
Total Reported CPU Utilization..

■ Max. Total C.. ■ Avg. Total C.. ■ Max. Cpu Li..



Total Memory Utilization..

■ Max. Total .. ■ Avg. Total M.. ■ Max. Total ..



Server Resource Management:
Explain how to use Logshark to manage server resources effectively, ensuring optimal utilization of CPU, memory, and disk space.

Logshark Analysis - Server Resource Manager

Server Resource Manager Reported Utilization By Process

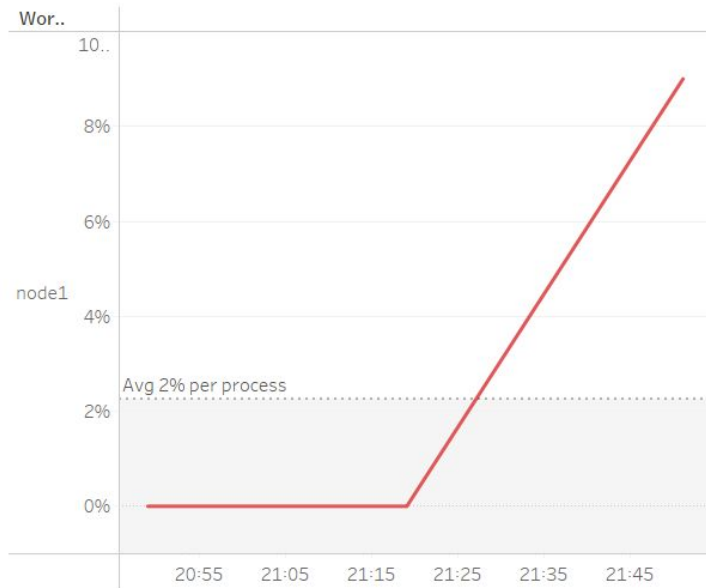
This dashboard shows Tableau Server Resource Manager logging activity on the input logset.

There are multiple reasons why a performance issue may not be visible [here](#). Only a few processes are actively monitored by Tableau Server. Log files may sometimes fail to be collected during periods of performance degradation. Additionally, the customer may have reconfigured or disabled SRM entirely, or be using a product vers...

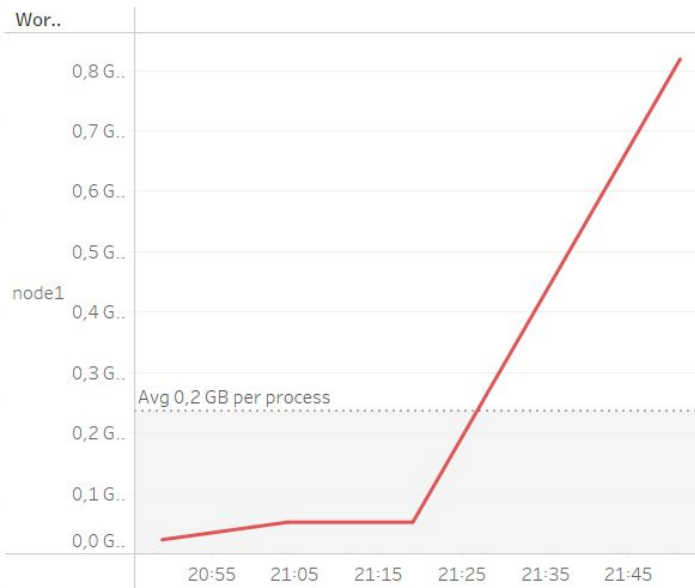
Timesta.. All values

Process ■ hyper

SRM Reported Utilization By Process



SRM Reported Memory Utilization By Process



Topology Adjustment

- Manual Changes:
Require a Tableau Server restart
- Example: Adding or removing a process other than Backgrounder or VizQL Server
- Dynamic Changes:
- Possible without restarting the server
- Example: Increasing or decreasing the number of Backgrounder or VizQL Server instances



Tableau Server Backup

Importance of Regular Backups

- **Data Integrity:** Regular backups ensure that your data is safe and can be restored in case of a failure.
- **Disaster Recovery:** Backups are crucial for disaster recovery, allowing you to quickly restore your server and minimize downtime.

Steps for Performing Backups

1. Full Backup
2. Database Backup
3. Settings Backup
4. Schedule Backups the TSM command line tool to schedule regular backups.

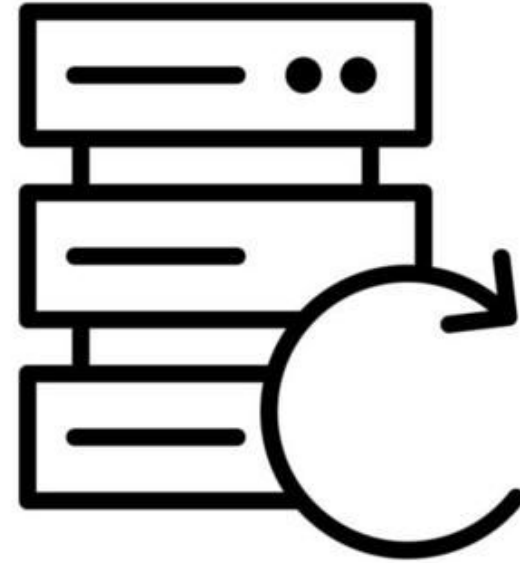


Tableau Server Backup

Best Practices for Backups

1. Secure Backup Files by storing them in an encrypted file system or a physically restricted location.
2. Copy the backup file to another computer, As a best practice, after the backup is created, copy the backup file to another location that is separate from Tableau Server.
3. Monitor Disk Space Regularly monitor disk space usage to ensure that backups do not consume excessive storage.

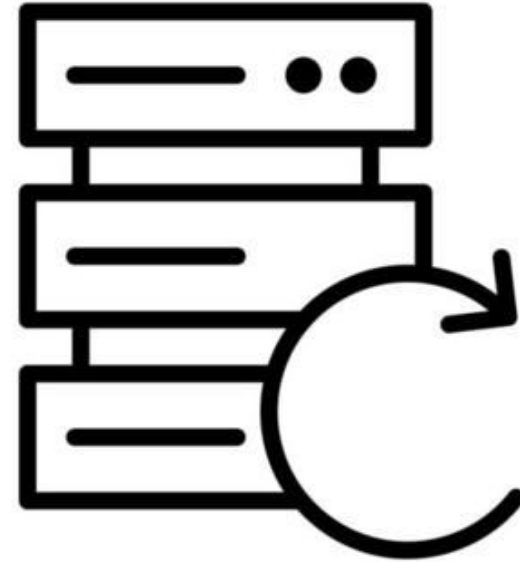


Tableau Server Cleanup

Regular Cleanup: Run the cleanup command regularly to maintain optimal performance.

Steps for Maintenance Cleanup

1. Archive Log Files
Use tsm maintenance ziplogs to archive log files.
Copy the ziplogs archive to a safe location.
2. Remove Unwanted Files
Use tsm maintenance cleanup to remove:
Log files older than seven days.
Temporary files.
3. Remove Other Files
Identify and delete:
Old backups from previous versions (.tsbak files).
Any other unnecessary files.



https://help.tableau.com/current/server/en-us/maintenance_cleanup.htm

Tableau Server Restore

- **Restore from Backup**
 - Use tsm maintenance restore to restore Tableau Server from a backup.
 - Ensure that the backup file is in the correct directory and that the Run As service account has read access to the backup file.
- **Compatibility**
 - Be aware of compatibility issues when restoring backups across different versions of Tableau Server.
 - Ensure that the backup file is compatible with the version of Tableau Server you are restoring to.

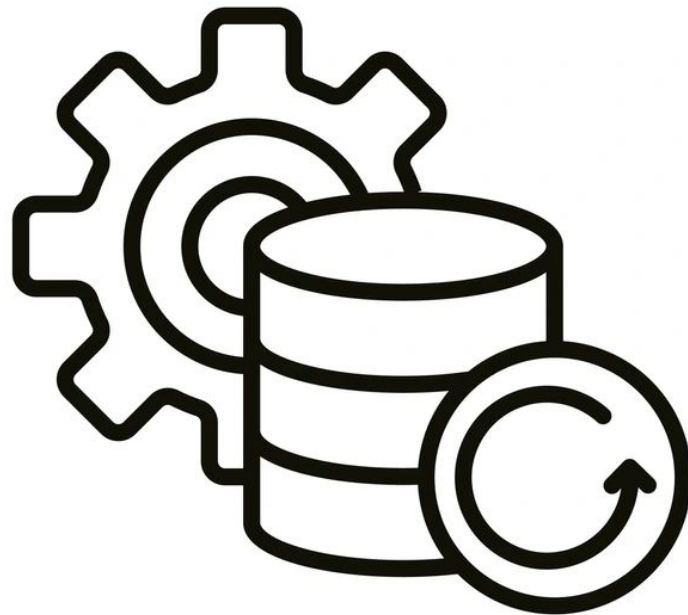


Tableau Server Upgrade

Upgrade Type

Current version	Upgrading to version	Install location	Upgrade procedure
<i>What version are you upgrading from?</i>	<i>If you aren't sure how to find your version, see Determine your Tableau Server version below.</i>	<i>If you do not know where Tableau Server is installed, see Determine your Tableau Server install location below.</i>	<i>Upgrade steps to use</i>
2018.2.x or later	2018.2.x or later	default or non-default location	Upgrading from 2018.2 and Later (Windows)
2018.1.x or earlier	2018.2.x or later	default or non-default location	Upgrade from Tableau Server 2018.1.x or Earlier  in the 2021.4 Server Help.

You must download and copy the Tableau Server Setup program to the computer running Tableau Server. If you are running a multi-node deployment of Tableau Server, copy the Setup program to each node in the cluster.

<https://help.tableau.com/current/server/en-us/upgrade.htm>

Tableau Server Upgrade Consideration

Consideration	Details
OS Compatibility	Ensure the OS is compatible with the current and target versions of Tableau Server.
Resources	Meet or exceed the minimum RAM, storage, and processor requirements for the chosen version.
Tableau Version	Determine the current and target versions using the Tableau Release Navigator.
Topology	Decide between a single node or high availability setup.
Upgrade Mechanism	Choose between blue/green upgrade (minimal downtime) or in-place upgrade (shut down server).

Suggested Tableau Server Maintenance Based on my Experience

1. **Implement the 3-2-1 Backup Strategy**

- a. Data Integrity: Ensure data safety with 3 copies of your data.
- b. Media Diversity: Store backups on two different media.
- c. Off-Site Backup: Maintain one copy off-site for disaster recovery.

2. **Automated Backups in Development Environment**

Development Environment: Automate backups and restores for quick recovery.

3. **Regular Health Checks and Log Shark Analysis**

- a. Health Checks: Perform CPU, memory, and disk space checks every 1-3 months.
- b. Log Shark Analysis: Manage server resources effectively to optimize performance.
- c. Cleanup: Run regular cleanup every 3 months to maintain optimal performance.

4. **Upgrade to a Stable Version**

Upgrade to the version one below the newest for more stable performance.

Disclaimer:

These suggestions may vary and are influenced by factors such as the number of workbooks, server specifications, and user behavior. Please follow the official Tableau Server guide for detailed instructions and best practices.

Automated Tableau Server Maintenance

 **tableau-server** Public

 Pin  Unwatch 1  Fork 0  Star 0

 main  1 Branch  0 Tags

 Add file

 Code

About

 Tableau Server Maintenance and Testing Guide

 Readme

 Activity

 0 stars

 1 watching

 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages



Language	Percentage
HTML	65.4%
Batchfile	34.6%

 **ikhsannur1996** Update README.md

8699e6b · last year 47 Commits

anti-virus	Update README.md	last year
tableau-backup	Create README.md	last year
tableau-maintenance	Update README.md	last year
testing-local-embed	Create README.md	last year
trusted-authentication	Add files via upload	last year
README.md	Update README.md	last year

 README

Tableau Server Maintenance and Testing Guide

Welcome to the Tableau Server Maintenance and Testing Guide. This comprehensive document will walk you through the essential steps for maintaining your Tableau Server, testing the embedding of Tableau visualizations, and verifying Trusted Authentication settings. Following these steps will help you ensure the smooth operation and integration of

<https://github.com/ikhsannur1996/tableau-server>

Reference

https://help.tableau.com/current/server/en-us/install_config_top.htm

<https://www.tableau.com/products/server>

<https://tableau.github.io/Logshark/>

https://help.tableau.com/current/server/en-us/maintenance_cleanup.htm

https://help.tableau.com/current/blueprint/en-us/bp_maintenance.htm

https://help.tableau.com/current/server/en-us/server_process_hot_topo.htm

<https://www.tableau.com/learn/webinars/establishing-server-maintenance-best-practices#video>

https://help.tableau.com/current/server/en-us/backup_restore.htm

<https://help.tableau.com/current/server/en-us/upgrade.htm>

<https://github.com/ikhsannur1996/tableau-server>